# Name the force and its source

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Identify forces on a chosen object and name each interaction partner.

**SUCCESS CRITERIA - I can...**

1. Name the chosen object and each force source.
2. Distinguish mass from weight with units.
3. Include only forces acting on the chosen object.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “interaction” and “weight” to explain your model or result. Sentence starters are optional.

**interaction**: two objects affecting one another

**weight**: the gravitational force on an object

**normal force**: a support force perpendicular to a contact surface

## 01 Bring back an earlier idea

Velocity changes from -4 to -24 m/s in 4 s, then stays at -24 m/s for 8 s.
**Find** acceleration and describe the speed trend in each interval.

**Word help:** acceleration = rate of change of velocity

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $F_g = mg$.
Use g = 9.8 N/kg unless stated. Normal force is perpendicular to contact; tension follows the string. Name the source.

**Worked reasoning:** A book rests on a table. Earth pulls the book down; the table pushes the book up. Each force label names both the force and the object producing it.

**Your turn: Draw** the book's force diagram.
**Name** the force on Earth paired with the book's weight.

**Sentence starter:** The ... in my model represents ... . Its label shows ... .

**Marzano check - circle:** ① ② ③

## 03 Inventory the interactions

A 5 kg crate is pulled by a rope across a rough level floor.
**Identify** every external force on the crate and the source of each. Do not assume force magnitudes without more information.

**Sentence starter:** I notice ... . This supports ... because ... .



Add and label force arrows.

The dot represents the object. Chosen object: crate

**Marzano check - circle:** ① ② ③

## 04 Mass and weight in two places

A 3.0 kg object is on Earth (g=9.8 N/kg), then on the Moon (g=1.6 N/kg).
**Find** each weight. What happens to mass?

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

### 05 Normal is not always weight

A 5 kg box on a level floor has no vertical acceleration. A rope pulls with a 12 N upward component.
**Calculate** the normal force and explain why it differs from mg.

**Sentence starter:** Both … . They differ in … because … .

**Marzano check - circle:** 1 2 3

### 06 Choose a method

A puck follows a circle at a steady 5 m/s. A student calls it equilibrium.
**Draw** velocity at two points and correct the claim.

**Sentence starter:** The … in my model represents … . Its label shows … .

**Marzano check - circle:** 1 2 3

### 07 Exit check

A 2.0 kg object hangs at rest from a vertical string.
**Name** the two forces, their sources, magnitudes and directions. Is object-on-string a force on the object?

**Word help:** at rest = not moving in the chosen reference frame

**Sentence starter:** I notice … . This supports … because … .

**Marzano check - circle:** 1 2 3

### 08 STRETCH: Apply it another way

A 2 kg book rests on a table while a hand pushes down with 15 N.
**Use** g=10 N/kg.
**Find** the normal force and draw the third-law partner of each force on its separate object.

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** 1 2 3